

Assignment 3

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Code 1: Find 2nd Lowest and 2nd Highest

```
#include <iostream>
using namespace std;

int calculate2Low(int arr[], int size) {
    int lowest = arr[0], secondLowest = arr[0];
    for (int i = 1; i < size; i++) {
        if (arr[i] < lowest) {
            secondLowest = lowest;
            lowest = arr[i];
        }
        else if (arr[i] < secondLowest && arr[i] != lowest) {
            secondLowest = arr[i];
        }
    }
    return secondLowest;
}

int calculate2High(int arr[], int size) {
    int highest = arr[0], secondHighest = arr[0];
    for (int i = 1; i < size; i++) {
        if (arr[i] > highest) {
            secondHighest = highest;
            highest = arr[i];
        }
        else if (arr[i] > secondHighest && arr[i] != highest) {
            secondHighest = arr[i];
        }
    }
    return secondHighest;
}

int main() {
    int n;
    cout << "Enter number of students: ";
    cin >> n;
    int scores[n];
    cout << "Enter scores:"
```

```

";
    for (int i = 0; i < n; i++) cin >> scores[i];
    cout << "2nd Lowest Score: " << calculate2Low(scores, n) << endl;
    cout << "2nd Highest Score: " << calculate2High(scores, n);
    return 0;
}

```

Code 2: Calculate Sum of Scores

```

#include <iostream>
#include <iomanip>
using namespace std;

double calculateSum(int arr[], int size) {
    double sum = 0;
    for (int i = 0; i < size; i++) sum += arr[i];
    return sum;
}

int main() {
    int n;
    cout << "Enter number of students: ";
    cin >> n;
    int scores[n];
    cout << "Enter scores:
";
    for (int i = 0; i < n; i++) cin >> scores[i];
    cout << fixed << setprecision(2);
    cout << "Sum = " << calculateSum(scores, n);
    return 0;
}

```

Code 3: Celsius to Fahrenheit

```

#include <iostream>
#include <iomanip>
using namespace std;

double celsiusToFahrenheit(double celsius) {
    return (celsius * 9.0 / 5.0) + 32;
}

int main() {
    double celsius;
    cout << "Enter temperature in Celsius: ";

```

```

    cin >> celsius;
    cout << fixed << setprecision(1);
    cout << "Temperature in Fahrenheit: " << celsiusToFahrenheit(celsius);
    return 0;
}

```

Code 4: Leap Year Checker

```

#include <iostream>
using namespace std;

bool isLeapYear(int year) {
    if ((year % 400 == 0) || (year % 4 == 0 && year % 100 != 0)) {
        return true;
    }
    return false;
}

int main() {
    cout << isLeapYear(2024);
    return 0;
}

```

Code 5: Count Vowels

```

#include <iostream>
using namespace std;

int countVowels(char word[]) {
    int count = 0;
    for (int i = 0; word[i] != '\0'; i++) {
        if (word[i] == 'a' || word[i] == 'e' || word[i] == 'i' || word[i] == 'o' || word[i] == 'u' ||
            word[i] == 'A' || word[i] == 'E' || word[i] == 'I' || word[i] == 'O' || word[i] == 'U') {
            count++;
        }
    }
    return count;
}

int main() {
    char word[100];
    cout << "Enter a word: ";
    cin >> word;
    cout << "Total Vowels: " << countVowels(word);
}

```

```
    return 0;
}
```

Code 6: Bubble Sort

```
#include <iostream>
using namespace std;

void bubbleSort(char arr[], int size) {
    for (int i = 0; i < size - 1; i++) {
        for (int j = 0; j < size - i - 1; j++) {
            if (arr[j] > arr[j + 1]) {
                char temp = arr[j];
                arr[j] = arr[j + 1];
                arr[j + 1] = temp;
            }
        }
    }
}

int main() {
    char letters[5];
    cout << "Enter 5 letters: ";
    for (int i = 0; i < 5; i++) cin >> letters[i];
    bubbleSort(letters, 5);
    cout << "Sorted Letters: ";
    for (int i = 0; i < 5; i++) cout << letters[i] << " ";
    return 0;
}
```